Therefore, the line $x=\frac{\pi}{2}$ is a vertical asymptote to the graph. Similarly, if x is approaching to $-\frac{\pi}{2}$, the ratio $\frac{\sin x}{\cos x}$ is negative and large in magnitude and thus, approaching to $-\infty$. So, the line $x=-\frac{\pi}{2}$ is also a vertical asymptote to the graph. Hence, we get a branch of the graph of $y=\tan x$ for $-\frac{\pi}{2}<x<\frac{\pi}{2}$ as shown in the Fig 4.15. The interval $\left(-\frac{\pi}{2},\ \frac{\pi}{2}\right)$ is called the **principal domain** of $y=\tan x$.

Since the tangent function is defined for all real numbers except at $x=(2n+1)\frac{\pi}{2}$, $n\in\mathbb{Z}$, and is increasing , we have vertical asymptotes $x=(2n+1)\frac{\pi}{2}$, $n\in\mathbb{Z}$. As branches of $y=\tan x$ are symmetric with respect to $x=n\pi, n\in\mathbb{Z}$, the entire graph of $y=\tan x$ is shown in Fig. 4.16.

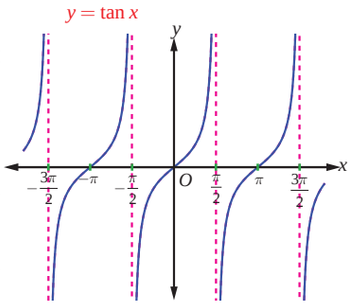


Fig. 4.16

Note

From the graph, it is seen that $y=\tan x$ is positive for $0<x<\frac{\pi}{2}$ and $\pi<x<\frac{3\pi}{2}$; $y=\tan x$ is negative for $\frac{\pi}{2}<x<\pi$ and $\frac{3\pi}{2}<x<2\pi$.

4.5.2 Properties of the tangent function

From the graph of $y=\tan x$, we observe the following properties of tangent function.

(i) The graph is not continuous and has discontinuity points at $x=(2n+1)\frac{\pi}{2},\ n\in\mathbb{Z}$.

(ii) The partial graph is symmetric about the origin for $-\frac{\pi}{2}<x<\frac{\pi}{2}$.

(iii) It has infinitely many vertical asymptotes $x=(2n+1)\frac{\pi}{2},\ n\in\mathbb{Z}$.

(iv) The tangent function has neither maximum nor minimum.

Remark

(i) The graph of $y=a\tan bx$ goes through one complete cycle for

$-\frac{\pi}{2|b|}<x<\frac{\pi}{2|b|}$ and its period is $\frac{\pi}{|b|}$.

(ii) For $y=a\tan bx$, the asymptotes are the lines $x=\frac{\pi}{2|b|}+\frac{\pi}{|b|}k,\ k\in\mathbb{Z}$.

(iii) Since the tangent function has no maximum and no minimum value, the term **amplitude** for $\tan x$ cannot be defined.

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